

Figure 1: List of nodes used for creating the decision tree

Designing the supervised ML car evaluation decision tree model

Here are the steps needed to create a decision tree model for car evaluation.

- Create a new workflow for the car evaluation model as seen in Figure 2.
- Drag the .csv file into KNIME to create and initialise the CSV reader node.
- Connect it to a colour manager, where each genre can be assigned a specific colour under *swatches* or a preset under *palettes*.
- Connect the colour manager to the *partitioning* node, which is used to separate the given data set into two partitions — one consisting of 80 per cent of the data and the other consisting of the remaining 20 per cent of the data. Thus the data is partitioned into two sets for testing and training data in an 80:20 ratio. This partition is done by choosing the *randomly* option.
- Connect the *partitioning* node to the decision tree learner. Configure it as a class column as units, and change the records as per the required analysis.
- Connect the decision tree learner to the decision tree predictor. Send the smaller partition (test data) to the decision tree learner in order to classify the values, which can then be predicted by the decision tree predictor.
- Finally, send the output to the *scorer* node to measure the performance metric score. Compile and execute all the nodes at each stage of connection to get proper outputs.
- Now, to view the output as a report, connect it to the PMML writer node.

Node configurations

CSV reader: This node reads CSV files. You can also use this node if the workflow is created in a server or batch environment, and where the structure of the input files changes between different invocations. The car evaluation data set is loaded using the CSV reader.

Colour manager node: Colours can be assigned for either nominal (possible values have to be available) or numeric columns (with lower and upper bounds). If these bounds are not available, a ‘?’ is provided as a minimum and maximum value. The values are then computed during execution. If a column attribute is selected, the colour can be

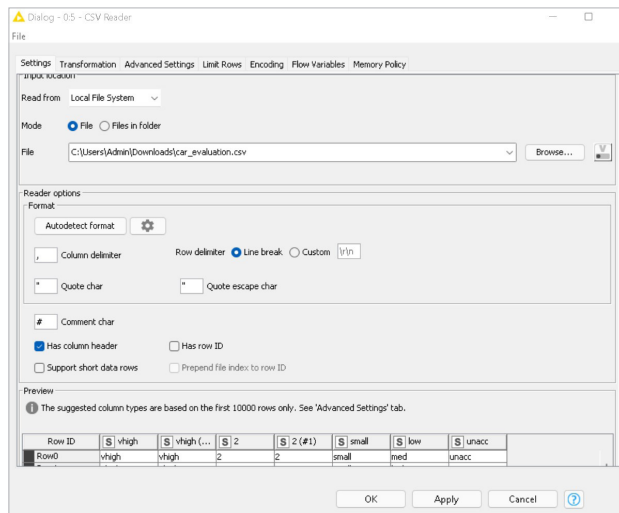


Figure 2: Configuration of CSV reader node

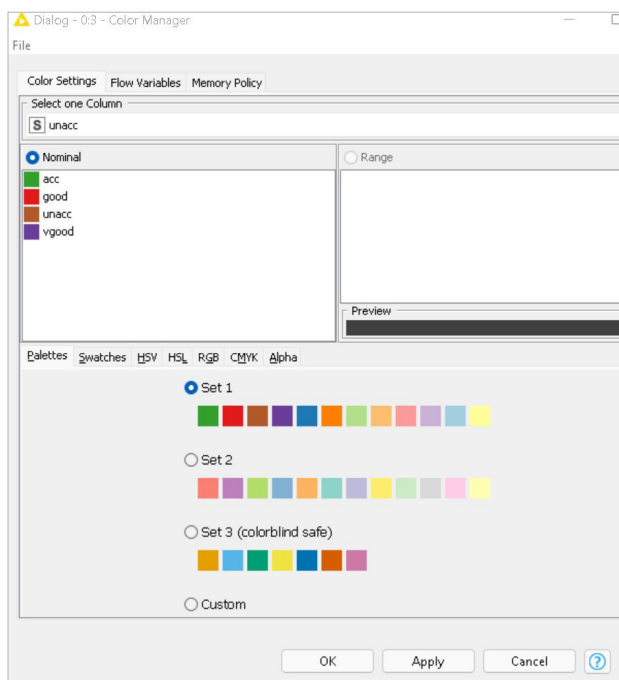


Figure 3: Configuration of colour manager node

changed with the colour chooser.

Partitioning node: The input table is split into two partitions (i.e., row-wise), e.g., train and test data. In this case, it is 80 per cent training and 20 per cent testing.

Decision tree learner node: This node induces a classification decision tree in the main memory. The target attribute must be nominal. The other attributes used for decision making can be either nominal or numerical. Numeric splits are always binary (two outcomes), dividing the domain into two partitions at a given split point. Here the target variable is ‘Evaluation’